

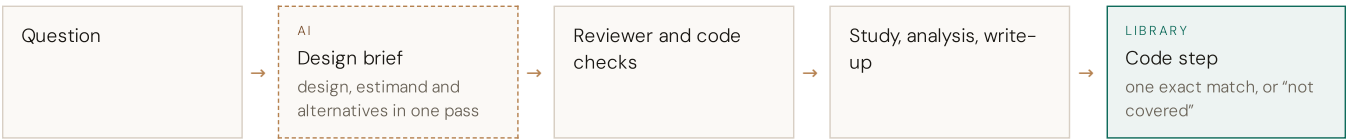
From a question to a *design*

Mallard is changing the order in which it builds a study plan. It will first pin down exactly what the question is asking for, as a typed estimand, and confirm it with the clinician. Rules and a tested code library then decide which analysis methods are valid. Only after that does the AI recommend a design and set out the alternatives.

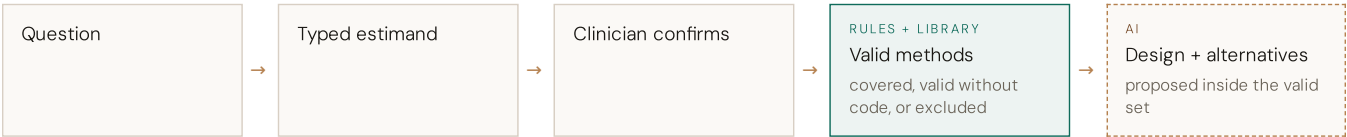
Proposed approach, in development. This page follows Mallard's draft design-selector and code-library specs from 23 September 2026, which have not been adopted yet. Today, Mallard's AI drafts the design first, and the code library is consulted only when the plan's code is written.

WHAT CHANGES

TODAY

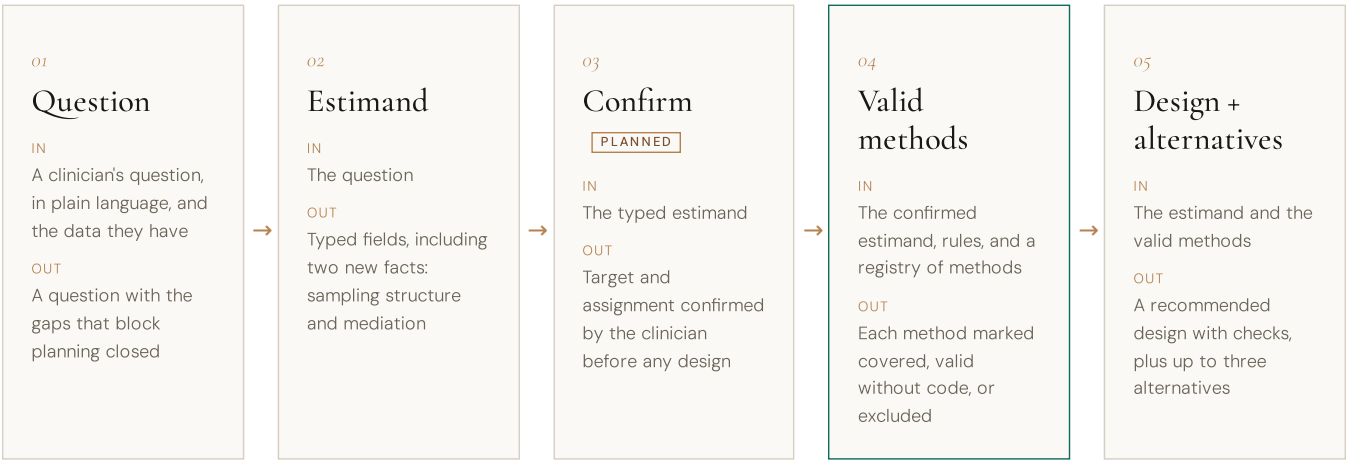


PROPOSED



Two things move. The code library goes from the last step to the middle, where it helps decide what is valid. The AI's design choice moves from the start to the end, so it works within limits the rules have already set. Dashed boxes are written by the AI; green boxes use the code library.

THE FIVE STAGES



The sections below follow one question through each stage. It is Mallard's own worked example. The library results are read from the metadata of the 72 entries the app ships today; the rules, registry and confirmation step are proposed.

01 Start with the question

A clinician states what they want to know and what data they have. They do not need to name a design or a method.

Does scheduling a follow-up appointment before discharge change 30-day unplanned readmission?

Data: discharge and scheduling records from a hospital medicine service.

WHAT HAPPENS HERE

- A readiness check lists what is still missing before a plan can be written. Anything the clinician has already answered is dropped from the list.
- Mallard asks sharpening questions only when the answer would change the plan. For example: *does the scheduling system record when an appointment was created, or only when it takes place?*

Define the estimand

The question is turned into an estimand: the exact quantity the study will estimate. Every field has a fixed vocabulary, so the rest of the pipeline can check it with code instead of interpreting prose.

target associational GATE	population Adults discharged alive from hospital medicine	exposure Follow-up scheduled before discharge	comparison No appointment scheduled before discharge
outcome Time to first unplanned readmission	outcomeType timeToEvent GATE	summaryMeasure hazardRatio GATE	timeHorizon 30 days after discharge
framing estimation	assignment nonrandomized	samplingStructure independent NEW GATE	repeatedMeasures false GATE
competingEvents true (death within 30 days) GATE	mediation false NEW		

☐ Shaded fields are checked against each method in stage O4. Fields tagged “new” are proposed additions to the intake.

WHY THIS COMES FIRST

- Competing events is a single yes/no field, and it changes which methods are valid. A standard Cox model treats a patient who died as simply censored, which overstates readmission.
- Sampling structure replaces a yes/no “clustered” flag with the code library’s own four values: independent, clustered, matched, or survey. That lets the rules see matched case-control and survey designs, which they cannot today.
- Mediation records whether the question asks how much of an effect works through a middle step. Today that is detected by searching the plan’s wording.

Every later rule trusts the typed facts, so a mistyped question gets confidently wrong answers. Before any design is proposed, the clinician confirms the facts that matter most.

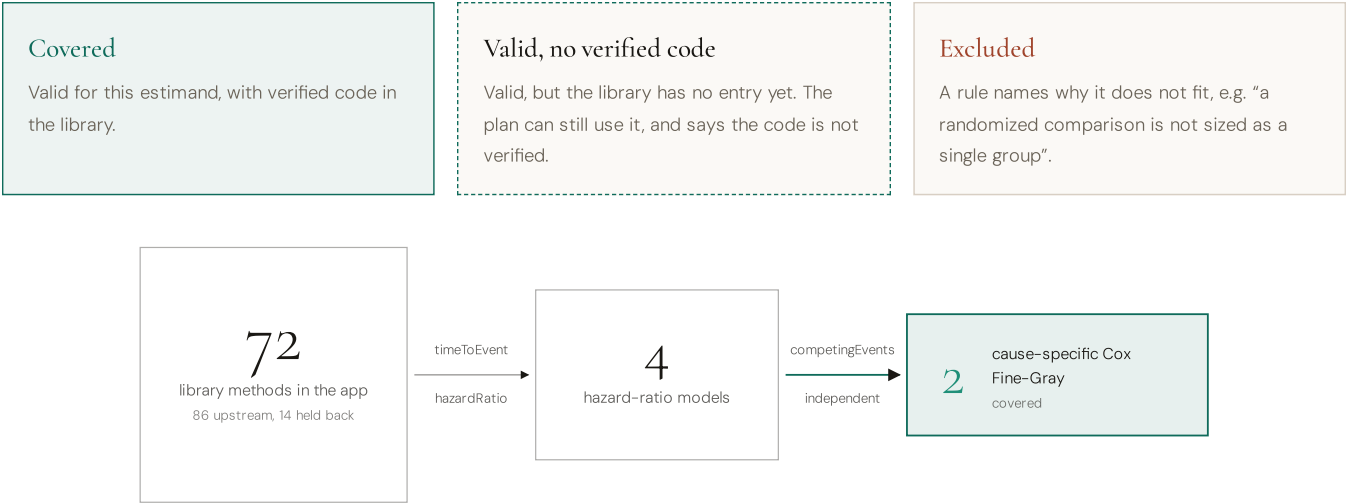
- Target: is this about a cause, an association, a description, or a prediction? *Here: an association.*
- Assignment: was the comparison randomized? *Here: no, the groups are observed.*
- Will each case be matched to specific controls (for example, on age and sex)? *Here: no.*
- Is part of the question how much of the effect works through a middle step? *Here: no.*

WHY IT IS NEEDED

- In Mallard's 23 September audit, 7 of 28 plans typed a fact differently from the reference answer, and one randomized trial was typed as descriptive three times.
 - "Not sure yet" is an allowed answer; it simply stops the related rules from firing. A value read from the description is shown back to the clinician, never applied silently.
 - This step is named in the draft spec but will be specified separately, so its exact form may change.
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04 Find the valid methods

Rules on the typed facts decide which analysis methods are valid. A registry lists every method, whether or not code exists for it. The code library is the verified part of that registry: its entries have R and/or Python code that has been run on a test dataset and checked against expected output. Each method ends up in one of three states.



Each arrow applies typed fields from the estimand. Outcome type and summary measure narrow 72 methods to 4 hazard-ratio models. Competing events then removes the standard Cox model and its time-varying variant, leaving 2 covered methods.

THE CHECK, FIELD BY FIELD

Library entry	target associational	outcomeType timeToEvent	summaryMeasure hazardRatio	competingEvents true	sampling independent	Result
Cause-specific Cox cause-specific-cox	✓	✓	✓	✓	✓	COVERED · R + PY
Fine-Gray subdistribution fine-gray-subdistribution	✓	✓	✓	✓	✓	COVERED · R
Cox proportional hazards cox-proportional-hazards	✓	✓	✓	✗ false only	✓	EXCLUDED
Cox with time-varying exposure cox-time-varying-exposure	✓	✓	✓	✗ false only	✓	EXCLUDED
Kaplan-Meier + log-rank kaplan-meier-logrank	✓	✓	✗ other	✗ false only	✓	EXCLUDED

Rules decide validity

Whether a method is valid depends on rules over typed facts, not on whether code exists. In the audit, when the library had 16 entries, only 17 of 75 plans matched any of them, and most of the rest were sound plans.

Exact, not nearest

Methods are not scored or ranked. The standard Cox model agrees on four of five fields and is still excluded, because it treats a death as an ordinary dropout.

Nothing is substituted

When no covered method fits, the plan says so. It does not quietly swap in a method for a different estimand that the library happens to have.

05 Recommend a design, show the alternatives

The AI proposes the recommended design and the alternatives, working inside the set of valid methods. The rules narrow the analyses and sample-size methods; choosing among designs is still the AI's job. Each option shows what it costs and what it would let you claim.

RECOMMENDED

ADJUSTED ASSOCIATION

MONTHS

EXISTING RECORDS

Retrospective cohort from discharge and scheduling records, with death as a competing risk

The exposure is already recorded at discharge, and the outcome is recorded in the same system. A cohort can therefore answer the question from data that already exists, without paying to collect new data. Because the two groups differ systematically, the analysis adjusts for confounders, and the result is reported as an association rather than an effect.

PRIMARY ANALYSIS · COVERED

COMPANION · COVERED

Cause-specific Cox regression, reported as an adjusted hazard ratio

Fine-Gray regression for the cumulative incidence of readmission

DESIGN-FIT CHECKS TO RUN BEFORE COMMITTING

1

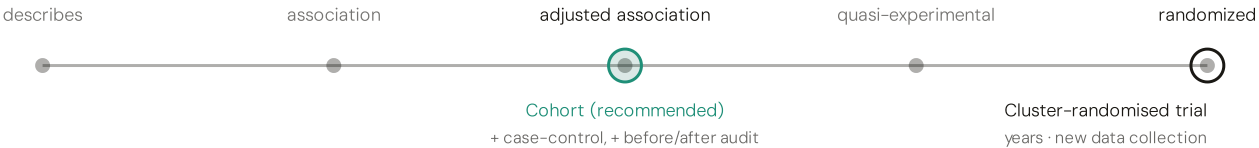
At least 95% of appointments have a creation time before the discharge time.
If not: restrict to reliable timestamps, or switch to the case-control design.

2

Deaths recorded in the system are within 10% of an external death count.
If not: report the cause-specific hazard only.

3

At least 10 readmissions per covariate in the adjustment set.
If not: widen the record window or reduce to a minimal sufficient set.



Three of the four designs support the same claim, an adjusted association. They differ in where the exposure data comes from and what can go wrong. Only allocating the change at random turns the result into a causal effect.

ADJUSTED ASSOCIATION

ABOUT A YEAR

Case-control, exposure abstracted from charts

WINS WHEN

The scheduling timestamp can't be trusted, and chart review is affordable for a sample of about 600 records but not for everyone.

ADJUSTED ASSOCIATION

MONTHS

Before-and-after audit around the scheduling change

WINS WHEN

The go-live date is known and records on both sides already exist. It describes a change around that date and makes no claim that the scheduling change caused it.

RANDOMIZED

YEARS

Cluster-randomised trial by ward

WINS WHEN

The scheduling change can be assigned to wards instead of observed. This is the only design here that can answer the question as a causal effect.

WHY THIS ORDER

When the design is chosen before the estimand is fixed, it is easy to reach for a familiar method that answers a slightly different question. Here, a standard Cox model would ignore deaths. Fixing and confirming the estimand first, and letting rules decide what is valid before the AI chooses, means the recommended design comes with methods that fit the question actually asked, and says plainly where verified code exists.

Status: the rules, the method registry, the two new intake facts and the confirmation step are drafts awaiting the owner's decisions. The first step proposed is to run the rules alongside plan generation, without changing any plan, until an audit shows they never exclude a method that expert reviewers accept.

Worked example adapted from [Ask Mallard](#) and its open code library.